Comparing Proportions with Odds Ratios

Presentation Outline

1. Population Odds Ratio
2. Sample Odds Ratio
3. Confidence Interval for Population Odds Ratio

Variables and Population Proportions

Variables:

- Grouping Variable ($I = 2$ groups)
- Response Variable ($J = 2$ categories)
    - Categories = Success and Failure

Population Proportions:

- p_1 = proportion of success in population 1.
- p_2 = proportion of success in population 2.

Comparing Proportions

Difference in Two Proportions: $p_1 - p_2$

Ignores Relative Size of p_1 and p_2:

p_1	p_2	$p_1 - p_2$	p_1/p_2
0.01	0.001	0.009	10
0.05	0.041	0.009	1.220
0.10	0.091	0.009	1.099
0.25	0.241	0.009	1.037
0.41	0.401	0.009	1.022

Population Odds

Ratio of proportion of success to proportion of failure in population.

$$\text{Odds} = \frac{p_i}{1 - p_i}$$

Interpretation:

- Odds $= 1$
  - $p_i = 1 - p_i$ implying $p_i = 0.5$
- Odds > 1
  - $p_i > 1 - p_i$ implying $p_i > 0.5$
- Odds < 1
  - $p_i < 1 - p_i$ implying $p_i < 0.5$

Population Odds Ratio

Ratio of population odds of success for group 1 to group 2.

$$\phi = \frac{\text{Odds for population 1}}{\text{Odds for population 2}} = \frac{\frac{p_1}{1-p_1}}{\frac{p_2}{1-p_2}}$$

Interpretation:

- $\phi = 1$
  - Odds of success is equal for the two groups, implying $p_1 = p_2$

Population Odds Ratio (cont.)

$$\phi = \frac{\frac{p_1}{1-p_1}}{\frac{p_2}{1-p_2}}$$

Interpretation:

- $\phi > 1$
  - The odds of success is larger for population 1 than for population 2, which implies $p_1 > p_2$
- $\phi < 1$
  - Odds of success is smaller for population 1 than for population 2 which implies $p_1 < p_2$.
- Note: ϕ does not indicate actual values of p_1 and p_2 - only relative values.

Data - Contingency Table

	Response Variable		
Group	Success	Failure	Total
1	Y_{11}	Y_{12}	n_1
2	Y_{21}	Y_{22}	n_2
Total	$Y_{.1}$	$Y_{.2}$	n

Sample Proportions

Define sample proportions as:

- $\hat{p}_1 = \frac{Y_{11}}{n_1}$ = observed proportion of successes in sample 1
- $1 - \hat{p}_1 = \frac{Y_{12}}{n_1}$ = observed proportion of failures in sample 1
- $\hat{p}_2 = \frac{Y_{21}}{n_2}$ = observed proportion of successes in sample 2
- $1 - \hat{p}_2 = \frac{Y_{22}}{n_2}$ = observed proportion of failures in sample 2

Sample Odds Ratio

Estimate Population Odds Ratio ϕ using these sample proportions.

$$\hat{\phi} = \frac{\frac{\hat{p}_1}{1-\hat{p}_1}}{\frac{\hat{p}_2}{1-\hat{p}_2}} = \frac{\frac{Y_{11}/n_1}{Y_{12}/n_1}}{\frac{Y_{21}/n_2}{Y_{22}/n_2}} = \frac{Y_{11}/Y_{12}}{Y_{21}/Y_{22}}$$

Confidence Interval for Population Odds Ratio

$\hat{\phi}$ will vary from sample to sample.

Estimate Population Odds Ratio ϕ with Confidence Interval

Two Steps:

- Calculate Confidence Interval for $\ln(\phi)$.
- Apply exponential function to both endpoints to obtain Confidence Interval for ϕ.

Confidence Interval for $\ln(\phi)$

Standard error of $\ln(\hat{\phi})$:

$$SE(\ln(\hat{\phi})) = \sqrt{\frac{1}{n_1\hat{p}_1} + \frac{1}{n_1(1-\hat{p}_1)} + \frac{1}{n_2\hat{p}_2} + \frac{1}{n_2(1-\hat{p}_2)}}$$

Confidence Interval:

$$\ln(\hat{\phi}) \pm z_{\alpha/2} * SE(\ln(\hat{\phi}))$$

where $z_{\alpha/2}$ is the upper $\alpha/2$ percentile of the standard Normal distribution.

Confidence Interval for ϕ

Take exponential function of endpoints to return values to original scale.

If CI for $\ln(\phi)$ is (a, b), then CI for ϕ is (e^a, e^b)

Example - Confidence Interval for ϕ

There has been debate among doctors over whether surgery can prolong life among men suffering from prostate cancer. In 2003, in a study published in the New England Journal of Medicine, men diagnosed with prostate cancer were randomly assigned to either undergo surgery or not. The response variable is whether or not the men died from prostate cancer.

Variables:

- Two Populations - Surgery, No Surgery
- Response Variable - Died from Prostate Cancer (Yes/No)

Example - Confidence Interval for ϕ

	Died from Prostate Cancer?		
Surgery	Yes	No	Total
Yes	16	331	347
No	31	317	348
Total	47	648	695

Example - Confidence Interval for ϕ

$$\hat{p}_1 = \frac{16}{347} \qquad 1 - \hat{p}_1 = \frac{331}{347} \qquad \hat{p}_2 = \frac{31}{348} \qquad 1 - \hat{p}_2 = \frac{317}{348}$$

$$\hat{\phi} = \frac{\frac{\hat{p}_1}{1-\hat{p}_1}}{\frac{\hat{p}_2}{1-\hat{p}_2}} = \frac{\frac{16/347}{331/347}}{\frac{31/348}{317/348}} = \frac{16/331}{31/317} = 0.4943$$

Example - Confidence Interval for ϕ

$$n_1\hat{p}_1 = 347\left(\frac{16}{347}\right) = 16 \qquad n_1(1-\hat{p}_1) = 347\left(\frac{331}{347}\right) = 331$$

$$n_2\hat{p}_2 = 348\left(\frac{31}{348}\right) = 31 \qquad n_2\left(1-\hat{p}_2\right) = 348\left(\frac{317}{348}\right) = 317$$

$$\begin{aligned} \mathrm{SE}(\ln(\hat{\phi})) &= \sqrt{\frac{1}{n_1\hat{p}_1} + \frac{1}{n_1(1-\hat{p}_1)} + \frac{1}{n_2\hat{p}_2} + \frac{1}{n_2(1-\hat{p}_2)}} \\ &= \sqrt{\frac{1}{16} + \frac{1}{331} + \frac{1}{31} + \frac{1}{317}} = 0.3177 \end{aligned}$$

Example - Confidence Interval for ϕ

95% CI for $\ln(\phi)$

$$\ln(\hat{\phi}) \pm z_{\alpha/2} * \mathrm{SE}(\ln(\hat{\phi})) = \ln(0.4943) \pm 1.96 * 0.3177 = (-1.3273, -0.0819)$$

Apply exponential function to both endpoints to get 95% CI for ϕ

$$(e^{-1.3273}, e^{-0.0819}) = (0.2651, 0.9214)$$

Example - Confidence Interval for ϕ

Interpretation:

We are 95% confident the population odds of dying from prostate cancer in the population of patients who have surgery is between 0.2651 and 0.9214 times the population odds of dying from prostate cancer in the population of patients who do not have surgery.

Relative Values:

Since both endpoints of CI are less than 1, this CI implies the proportion of patients in the population that have surgery that die from prostate cancer is less than the proportion of patients in the population who do not have surgery that die from prostate cancer.